$(s-1)$-Edge-Fault-Tolerant (Strongly) Hamiltonian Laceability in Exchanged Hypercube

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Abstract

The exchanged hypercube $EH(s,t)$ introduced by Loh et al., is obtained by systematically removing links from a hypercube. This paper considered the edge-fault-tolerant capability with respect to hamiltonian laceability of the exchanged hypercube. We showed that exchanged hypercubes have hamiltonian laceability and strongly hamiltonian laceability when the number of faulty edges are no more than $(s-1)$. That is the exchanged hypercube $EH(s,t)$ is $(s-1)$-edge-fault-tolerance with respect to hamiltonian laceability.

Keywords. Exchanged hypercubes, Edge-fault-tolerant, Hamiltonian laceability, Strongly hamiltonian laceability.

1 Introduction

An interconnection network can be represented by an undirected graph where the vertices represent the nodes and the edges represent the links. Let $G = (V, E)$ be a graph with vertex set V and edge set E. Two vertices u and v are *adjacent* if $(u, v) \in E$. A *uv-path* $P(u, v) = (u = u_1, u_2, ..., u_k = v)$ is a sequence of vertices such that the consecutive vertices are adjacent. A path in a graph G containing every vertex of G is called a *hamiltonian path.* A graph G is *hamiltonian-connected* if for every pair of distinct vertices u and v, there exists a hamiltonian uv-path. A *cycle* $(u_1, u_2, ..., u_k, u_1)$ is a closed path. A cycle containing every vertex of G is called a *hamiltonian cycle.* A graph G is *hamiltonian* if it contains a hamiltonian cycle.

When we discuss the hamiltonicity of a graph G, it is an important issue to investigate if G is hamiltonian or hamiltonian connected. For any bipartite graph with partite sets V_1 and V_2 such that $|V_1| \leq |V_2|$, there is no hamiltonian path between two vertices in $|V_1|$, bipartite graphs are not hamiltonian-connected. Simmon [12] introduces the concept of hamiltonian laceable for those hamiltonian bipartite graphs. A bipartite graph is *hamiltonian laceable* if there is a hamiltonian path between any two vertices in distinct partite sets. Hsieh et al. [4] further extend this concept into strongly hamiltonian laceable. A hamiltonian laceable graph $G = (V_1 \cap V_2, E)$ is *strong* if there is a path of length $|V_1 \cup V_2| - 2$ between any two vertices of the same partite set.

The edge fault-tolerant hamiltonicity, proposed by Hsieh, Chen, and Ho [3], measures the performance of the hamiltonian property in the faulted networks. Let F_e denote the set of faulty edges. A hamiltonian graph G is *k-edge fault-tolerant hamiltonian* if $G-F_e$ remain hamiltonian for every $F_e \subset E(G)$ with $|F_e| \leq k$. A hamiltonian laceable graph G is *k-edge fault tolerance* if $G - F_e$ remains hamiltonian laceable for every $F_e \subset E(G)$ with $|F_e| \leq k$.

The hypercube Q_n with recursive construction is one of the most popular interconnection network [5]. Hypercubes, which have many properties such as regularity, symmetricity and connectivity. A n-dimensional hypercube Q_n has 2^n vertices, which labeled by n-bit binary strings, and 2^{n-1} edges. The vertex of a hypercube Q_n can be represented as $V = \{b_n b_{n-1} \dots b_i \dots b_1 | b_k \in \{0, 1\}$ for $1 \leq k \leq n\}$. Two vertices of Q_n are adjacent if and only if their binary string differ in exactly one bit position. Harary[2] and Tsai[13] have discussed the laceability and fault-tolerance of laceability in hypercubes, respectively. There are many hypercube-like graphs, such as twisted cubes TQ_n, crossed cubes CQ_n, möbius cubes MQ_n, and exchanged hypercube $EH(s,t)$. These structures keep most of the properties from hypercubes.

The exchanged hypercube $EH(s,t)$ was proposed by Loh et al.[7] in 2005. It does not only keeps many properties from hypercube but also has relatively low link complexity. The number of vertices in $EH(s,t)$ is the same as a $(s+t+1)$-dimensional hypercube, but $EH(s,t)$ has much fewer edges than an $(s+t+1)$-dimensional hypercube does. Since the exchanged hypercubes have lots of advantages, there are some discussions about exchanged hypercubes [6] [7] [9] [10] [11].

According to [10], the vertex of an exchanged hypercube $EH(s,t)$ can be represented as

$$V = \{b_{s+t+1}b_{s+t}\dots b_i \dots b_{t+1}b_t \dots b_j \dots b_1 | \\ b_k \in \{0,1\} \text{ for } 1 \leq k \leq s+t+1\}$$

Let $v[x:y]$ denote the bit pattern of v from dimension y to dimension x and $H(x,y)$ denote the Hamming distance between x and y. Let $v[i]$ be the i-th bit of v which counts from right to left. The edge set E is composed of three disjoint subsets E_1, E_2, E_3 as follows:

$E_1 = \{(v_1,v_2) \in V \times V | v_1[s+t:1] = v_2[s+t:1]$, $v_1[s+t+1] \neq v_2[s+t+1]\}$, which connect the vertices differ by the last bit.

$E_2 = \{(v_1,v_2) \in V \times V | v_1[t:1] = v_2[t:1]$, $H(v_1[s+t:t+1], v_2[s+t:t+1]) = 1$, $v_1[s+t+1] = v_2[s+t+1] = 1\}$, which are the links between vertices with last bit 1.

$E_3 = \{(v_1,v_2) \in V \times V | v_1[s+t:t+1] = v_2[s+t:t+1], H(v_1[t:1], v_2[t:1]) = 1$, $v_1[s+t+1] = v_2[s+t+1] = 0\}$, which are the links between vertices with last bit 0.

Clearly, there are 2^{s+t+1} vertices and $(s+t+2)2^{s+t-1}$ edges in $EH(s,t)$. The illustration of $EH(1,2)$, $EH(1,3)$ and $EH(2,2)$ are shown in Figure 1.

[8] shows that $EH(s,t)(s \leq t \leq 2)$ is hamiltonian laceable and strongly hamiltonian laceable. This paper further considers the fault tolerance capacity of $EH(s,t)$ with respect to hamiltonian laceable and strongly hamiltonian laceable.

2 Main result

The neighborhood of vertex u is denoted by $N(u) = \{v \mid (u,v) \in E(G)\}$. The *degree of* u is defined as $\deg u = |N(u)|$. The *distance* $d(u,v)$ between two vertices u and v in G is the length of the shortest uv-path. We use $N_k(u) = \{v \mid d(u,v) = k\}$ to denote the *distance k neighborhood* of u for $k \geq 2$. Let $S \setminus T$ denote the complement of set T in set S. For the graph definitions and notations which are not mentioned here, we follow [1].

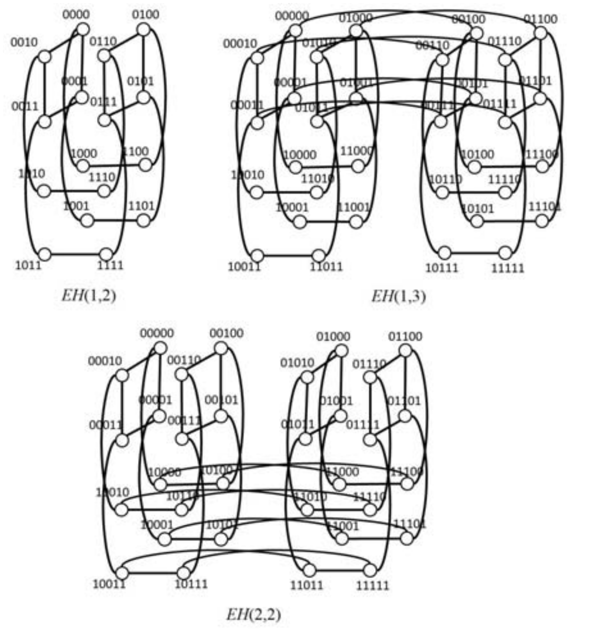

Figure 1: The example of $EH(1,2)$, $EH(1,3)$ and $EH(2,2)$

Property 1 (See [7]) *$EH(s,t)$ is isomorphic to $EH(t,s)$.*

Property 2 (See [7]) *$EH(s,t)$ can be decomposed into two copies of $EH(s-1,t)$ or two copies of $EH(s,t-1)$.*

Property 3 (See [9]) *For $s \leq t$, at least $s+1$ vertices (resp. $s+1$ edges) of $EH(s,t)$ are removed to get a disconnected graph.*

Since property 1 holds, we can always assume that $s \leq t$ when considering faulty-edges in $EH(s,t)$.

According to Property 2, we can decompose $G = EH(k+1,t)$ into two subgraphs: $L \cong R \cong EH(k,t)$, where the edges connect L and R belong to $E_2(G)$, and

$$V(L) = \{b_{k+t+1}b_{k+t}\dots b_i \dots b_{t+2}0b_t \dots b_j \dots b_1 | \\ b_p \in \{0,1\} \text{ for } 1 \leq p \leq k+t+1, \\ p \neq t+1\}$$
$$V(R) = \{b_{k+t+1}b_{k+t}\dots b_i \dots b_{t+2}1b_t \dots b_j \dots b_1 | \\ b_p \in \{0,1\} \text{ for } 1 \leq p \leq k+t+1, \\ p \neq t+1\}$$

Lemma 1 (from [10]) *Let $G = EH(s,t)$ be an exchanged hypercube graph for some $s,t \geq 1$. Then every vertex in $V_x = \{a_s \dots a_1 b_t \dots b_1 x | a_i, b_j \in \{0,1\},\ i \in [1,s],\ \text{and}\ j \in [1,t]\}$ for $x \in \{0,1\}$ are vertex-transitive.*

Lemma 2 (from [10]) *Let E_k^i and E_k^j of E_k ($k \in \{2,3\}$) be the edge subsets of $EH(s,t)$ with dimension i and j respectively. Then there is automorphism σ of $EH(s,t)$ such that $\sigma(E_k^i) = E_k^j$ and $\sigma(E_k^j) = E_k^i$.*

Lemma 3 (from [8]) *$EH(s,t)(s \leq t \leq 2)$ is hamilitonian laceable and strongly hamilitonian laceable.*

Let $V(EH(s,t))$ be decomposed into two partite sets V_1 and V_2. Through a brute force search by computer program, we are able to find a hamilitonian path from every vertex $u \in V_1$ to every vertex $v \in V_2$ and a path of length $|V_1 \cup V_2| - 2$ from each vertex $u \in V_i$ to each vertex $v \in V_i$ where $i \in \{1,2\}$. Hence we have Lemma 4.

Lemma 4 *Let F_e be the faulty edge set of $G = EH(2,2)$ and $|F_e| = 1$. $G - F_e$ is hamilitonian laceable and strongly hamilitonian laceable.*

Theorem 1 *Let F_e be the faulty edge set of $G = EH(s,t)(2 \leq s \leq t)$ and $|F_e| \leq s-1$. That is, $G - F_e$ is $(s-1)$-edge-fault-tolerance respect to hamilitonian laceable.*

Proof. Let $G' = G - F_e$. Since $|F_e| \leq s-1$, by property 3, G' is connected. Let V_1 and V_2 be two partite sets of $V(G')$. Since $EH(s,t)$ is isomorphic to $EH(t,s)$ (by Property 1). We just have to show the result by induction on s. For $s = t = 2$, by Lemma 3 and Lemma 4, we have $EH(2,2)$ is 1-edge fault tolerant with respect to hamilitonian laceable. Assume that for $2 \leq k \leq t$, when $|F_e| \leq k-1$, $EH(k,t)$ is hamilitonian laceable. For $s = k+1 \geq 3$ and $|F_e| \leq k$, we want to show that $EH(k+1,t)$ is hamilitonian laceable.

Since $|F_e| \leq k$, consider the worst case where $|F_e| = k$, and all of the faulty edges incident to a vertex u with degree $k+1$ in subgraph L. Then the vertex u will have only one non-faulty edge in L. By Lemma 2, we may use another dimension edge in E_2 to reconstruct the isomorphic graph $EH(k,t) \cong L \cong R$ so that the number of faulty edges in L' and R' are both less than k (see Figure 2).

Without loss of generality, let $u \in V_1$ and $v \in V_2$. We need to find a hamiltonian path between u and v. Consider following two subcases:

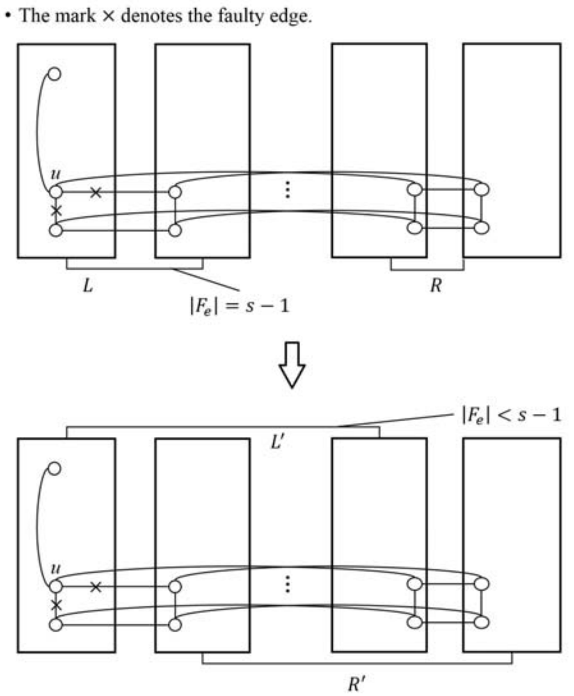

Figure 2: Reconstruct L' and R' so that there is a faulty edge connecting L' and R'.

Case 1. $u \in V(L')$, $v \in V(R')$. Since $|F_e| \leq k \leq 2^{k+t-2}$, there exists a vertex $u' \in V(L') \cap V_2$ incident with a non-faulty edge e connecting L' and R'. Let $v' \in V(R') \cap V_1$ be the vertex in R' incident with e. By inductive hypothesis, there is a hamiltonian uu'-path P_1 in L' and a hamiltonian $v'v$-path P_2 in R'. Then the path $P = P_1 + u'v' + P_2$ is a hamiltonian $uv-$path in $EH(k+1,t)$, which is illustrated in Figure 3.

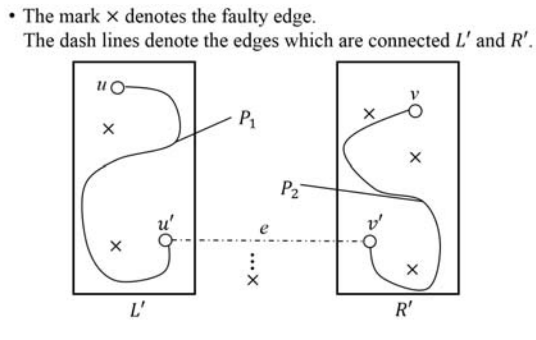

Figure 3: The example of Theorem 1 Case 1.

Case 2. u and v are both in $V(L')$ or both in $V(R')$. Without loss of generality, let $\{u,v\} \subseteq$

$V(L')$. By inductive hypothesis, there is a hamiltonian uv-path P' in L'. Since there are 2^{k+t} vertices in L' which incident which edges in E_2, hence $|P' \cap E_2| \geq 2^{k+t-1} - 1$. Since $|F_e| < k \leq 2^{k+t-2} \leq 2^{k+t-1} - 1$, there must exist an edge $u'v' \in P'$ such that u' and v' incident with non-faulty edges e_1 and e_2, which are some edges connecting L' and R', respectively. Let u'' and v'' be the vertices in R' which incident with e_1 and e_2, respectively. By inductive hypothesis, there is a hamilitonian $u''v''-$path P'' in R'. Let P_1 and P_2 denote the subpaths $u-u'$ and $v'-v$ in P', respectively. Then the path $P = P_1 + u'u'' + P'' + v''v' + P_2$ is a hamiltonian $uv-$path in $EH(k+1, t)$, which is illustrated in Figure 4.

By above cases we have G is $(s-1)$-edge-fault tolerance with respect to hamiltonian laceable. ■

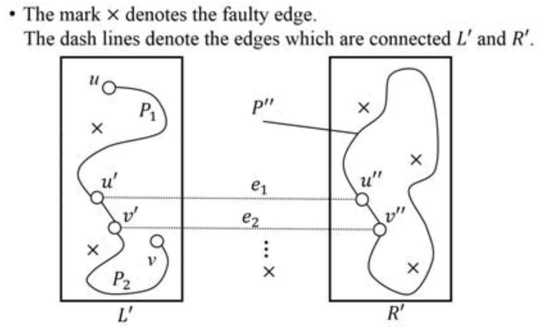

Figure 4: The example of Theorem 1 Case 2.

Theorem 2 *Let F_e be the faulty edge set of $G = EH(s,t)(2 \leq s \leq t)$ and $|F_e| \leq s-1$. That is, $G - F_e$ is $(s-1)$-edge-fault-tolerance respect to strongly hamilitonian laceable.*

Proof. We show the result by inductive on s similar using proof of Theorem 1. Let V_1 and V_2 be two partite sets of $G' = G - F_e$. By the same reason as in the proof of Theorem 1, we may decompose G into $L' \cong R' \cong EH(k,t)$ for some k, $2 \leq k \leq t$, such that $|F_e \cap E(L')| < k$ and $|F_e \cap E(R')| < k$. We have to construct a uv-path of length $|V(G)| - 2$ for u and v are in the same partite set. Without loss of generality, let $\{u, v\} \subseteq V_1$. Consider following two subcases:

Case 1. $u \in V(L'), v \in V(R')$. Let $u' \in V(L') \cap V_2$ such that u' incident with a non-faulty edge e connecting L' and R'. By Theorem 1, there is a hamiltonian uu'-path. Let v' be the vertex in $V(R')$ incident with e. Notice that $v' \in V(R') \cap V_1$. By inductive hypothesis, there is a $v'v$-path P_2 of length $|V(R')| - 2$. Then $P = P_1 + u'v' + P_2$ is a uv-path of length $|V(L')| - 1 + |V(R')| - 2 + 1 = |V(G)| - 2$ in $EH(k+1, t)$.

Case 2. u and v are both in $V(L')$ or both in $V(R')$. Without loss of generality, let $\{u, v\} \subseteq V(L')$. By induction hypothesis, there is a uv-path P' in L' with length $|V(L')| - 2$. For the same reason as in the proof of Theorem 1, there must exist an edge $u'v' \in P'$ such that u' and v' incident with non-faulty edges e_1 and e_2, which are some edges connecting L' and R', respectively. Let u'' and v'' be the vertices in R' which incident with e_1 and e_2, respectively. Since $(u', v') \subseteq E(L')$, we have $(u'', v'') \subseteq E(R')$, that is u'' and v'' are in distinct partite sets. By Theorem 1, there is a hamiltonian $u''v''$-path P'' in R'. Let P_1 and P_2 denote uu'-subpath and $v'v$-subpath pf P', respectively. Then the path $P = P_1 + u'u'' + P'' + v''v' + P_2$ is a uv-path of length $|V(L')| - 3 + |V(R')| - 1 + 2 = |V(G)| - 2$ in $EH(k+1, t)$.

By above cases we have G is $(s-1)$-edge-fault tolerance with respect to strongly hamiltonian laceable. ■

3 Conclusion

In [10], it showed that the exchanged hypercube may preserve the laceable capability which only about half edges of hypercubes. This paper further showed that the laceable property can be preserved in exchanged hypercubes even with $(s-1)$ faulty edges. That is exchanged hypercubes are $(s-1)$-edge-fault-tolerance respect to hamiltonian laceable and strongly hamiltonian laceable.

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